

Employment

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| <b>Graduate Research Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>NC State University</b>  | <b>Apr 2025 – Present</b>  |
| <ul style="list-style-type: none"><li>Designed and deployed a cloud-native ML data platform on GCP (Cloud Run) integrating JBrowse 2 genome browser, enabling feature extraction and model training on 500GB+ genomic datasets for 100+ researchers.</li><li>Improved ML model accuracy by 12% through systematic hyperparameter tuning and time-series feature engineering on biological sensor data; built scikit-learn evaluation pipelines with cross-validated metrics.</li></ul> |                             |                            |
| <b>Product Engineer Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Isomer AI</b>            | <b>May 2025 – May 2026</b> |
| <ul style="list-style-type: none"><li>Deployed 10+ production AI features into enterprise workflows: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows with MCP tooling, and Auth0 RBAC; 923 commits, 7 deployments.</li><li>Owned the full backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry APM); managed CI/CD pipelines, resolved production incidents on-call, and upheld SLA across enterprise client environments.</li></ul>          |                             |                            |
| <b>Software Developer Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Mphasis (HP Company)</b> | <b>Jun 2024 – Aug 2024</b> |
| <ul style="list-style-type: none"><li>Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML models, improving document processing accuracy by 25% across high-volume enterprise insurance pipelines.</li><li>Automated AWS and GCP deployments via CI/CD pipelines, integrated Grafana dashboards for real-time monitoring, and led model red-teaming to identify failure modes and improve production reliability.</li></ul>                   |                             |                            |
| <b>AI/ML Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Capgemini</b>            | <b>Jun 2023 – Aug 2023</b> |
| <ul style="list-style-type: none"><li>Led full SDLC for a data-driven support ticket analysis system using NLP classification models, reducing average resolution time by 15% and improving first-contact resolution rates across enterprise clients.</li><li>Built distributed data pipelines using Kafka streaming for real-time ingestion and Spark + Hadoop for batch processing, cutting ticket-processing latency by 20% and improving overall data throughput.</li></ul>        |                             |                            |

Education

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| <b>Raleigh, NC</b>                                                                                                                                                                                                                                                                             | <b>North Carolina State University</b> | <b>Aug 2024 – May 2026</b> |
| <ul style="list-style-type: none"><li>Master of Science in Computer Science</li><li>Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.</li></ul> |                                        |                            |
| <b>Mumbai, India</b>                                                                                                                                                                                                                                                                           | <b>Mumbai University</b>               | <b>Aug 2020 – May 2024</b> |
| <ul style="list-style-type: none"><li>Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS &amp; Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.</li></ul>                                         |                                        |                            |

Projects

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| <b>AlgoPulse</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Agentic AI Orchestration Platform</b> | <b>2025</b> |
| <ul style="list-style-type: none"><li>Built an event-driven AI orchestration platform in TypeScript with Temporal.io workflows and Kafka; identified API-polling bottleneck under load, pivoted to Supabase Edge Functions; scaled to 2,000 async req/s.</li><li>Built Temporal.io + Kafka worker pipeline for state sync and fault tolerance; shipped React Native mobile client with real-time AI inference; handled 2,000 concurrent requests with zero state loss.</li></ul>    |                                          |             |
| <b>CortexQ</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Python LLM Orchestration Platform</b> | <b>2025</b> |
| <ul style="list-style-type: none"><li>Built a Python FastAPI microservice with EWMA load balancing across Claude, GPT-4o, and Gemini backends; Redis request queue, Kubernetes auto-scaling via KEDA; 88 pytest tests with full CI integration.</li><li>Implemented a Python Kopf CRD operator reconciling Kubernetes resources; wired OpenTelemetry + Prometheus + Grafana for production observability; shipped GitOps via Helm and ArgoCD with zero-downtime rollouts.</li></ul> |                                          |             |

Languages and Technologies

- Languages:** TypeScript, JavaScript, Python, Java, SQL, Ruby
- AI/ML:** LLMs (Claude, GPT-4o, Gemini), MCP tooling, Temporal.io, RAG, embeddings, PyTorch, scikit-learn
- Frameworks:** Node.js, NestJS, Next.js, React, React Native, Express, FastAPI
- Cloud & Infra:** AWS, GCP, PostgreSQL, Redis, Docker, Kubernetes, Terraform, OpenTelemetry, GitHub Actions

Additional Experience and Awards

- Open Source:** Contributed to Expertiza Rails platform (5,000+ users, 300+ contributors); added 110+ RSpec tests, increased coverage by 25%, reduced redundancy by 35%.
- Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; shipped 3 full-stack MVPs end-to-end within 48-hour competitive sprints.
- Publications:** Co-authored CV-Based Cybersecurity Threat Detection (Springer 2023) and Mapify (IEEE 2022); presented to 500+ global attendees.
- GDSC Lead:** Led Google Developer Student Club; organized 4+ tech events (hackathons, conferences) reaching 500+ students.